

lapa-v0.1.2-instruct EN: religion — every mitigation against the baseline on the same rows

| Mitigation               | Condition | Unstable sets %<br>[95% CI] | Δ pp   | Fixed :<br>broken | p     | MAD<br>pp ↓ | AR range<br>pp ↓ | Cohen's h<br>↓ | Inconsist.<br>% ↓ | Hire<br>%     | Utility<br>% ↑ | Mean<br>FS  | Attr.<br>mentioned % |
|--------------------------|-----------|-----------------------------|--------|-------------------|-------|-------------|------------------|----------------|-------------------|---------------|----------------|-------------|----------------------|
| baseline (no mitigation) | implicit  | 23.78 [19.56, 28.44]        |        |                   |       | 2.94        | 13.11            | 0.07           | 5.19              | 73.14         | 55.06          | 0.57        | 1.09                 |
| SFT (LoRA, merged)       | implicit  | 1.78 [0.67, 3.11]           | -22.00 | 107 : 8           | 3e-23 | 0.22 (2.94) | 0.89 (13.11)     | 0.01 (0.07)    | 0.54 (5.19)       | 17.80 (73.14) | 80.96 (55.06)  | 0.63 (0.57) | 0.00 (1.09)          |
| baseline (no mitigation) | explicit  | 18.89 [15.11, 22.89]        |        |                   |       | 3.19        | 13.33            | 0.14           | 4.12              | 75.14         | 54.20          | 0.57        | 0.47                 |
| SFT (LoRA, merged)       | explicit  | 0.22 [0.00, 0.67]           | -18.67 | 85 : 1            | 2e-24 | 0.04 (3.19) | 0.22 (13.33)     | 0.00 (0.14)    | 0.02 (4.12)       | 17.36 (75.14) | 80.47 (54.20)  | 0.63 (0.57) | 0.00 (0.47)          |

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate–job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

\* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.